Ch. 17

(37.) Lightning strike: $\Delta t = 45\ ms = 45 \times 10^{-3}\ sec$

$\Delta Q = 350\ C$

$$\rightarrow I = \frac{\Delta Q}{\Delta t} = \frac{350\ C}{45 \times 10^{-3}\ s} = \boxed{7.78 \times 10^{3}\ amp}$$

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(41) Platinum, room temp resistance $R_0 = 58.5\ \Omega$

$\rho = 1.06 \times 10^{-7}\ \Omega \cdot m$ [resistivity]

$\alpha = 3.9 \times 10^{-3}\ \frac{1}{^\circ C}$ [temp coefficient]

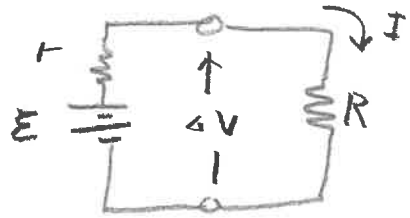
$$\rho = \rho_0 [1 + \alpha (T - T_0)]$$

$$\rightarrow R = R_0 [1 + \alpha (T - T_0)]$$

$\Delta T = T - T_0 = 0.1^\circ C$

$$\Delta R = R - R_0$$
$$= R_0 \alpha (T - T_0)$$
$$= R_0 \alpha \Delta T$$
$$= \boxed{0.0228\ \Omega}$$

(47.)



$\varepsilon = 18$ volts

$\Delta V = 17.4$ volts

$R = 130\ \Omega$

Determine internal resistance:

$$0 = \varepsilon - Ir - IR$$
$$= \varepsilon - I(r + R)$$

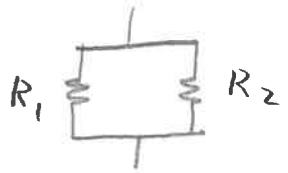
$$I = \frac{\Delta V}{R} = \boxed{0.134\ Amp}$$

$\rightarrow$ Thus, $\varepsilon = I(r + R)$

$$\rightarrow r + R = \varepsilon / I$$

$$r = \varepsilon / I - R = \frac{18\ volts}{0.134\ A} - 130\ \Omega = \boxed{4.48\ \Omega}$$

(53.)



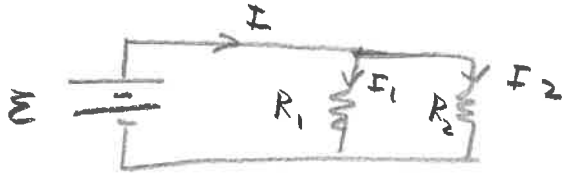
$\equiv$



$R_1 = 250\,\Omega$
$R_2 = 370\,\Omega$

a) $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} \rightarrow R_p = \frac{1}{\left(\frac{1}{R_1} + \frac{1}{R_2}\right)} = \boxed{149\,\Omega}$

b)



$\varepsilon = 12\ V$

$I = \frac{\varepsilon}{R_p} = \frac{12\ V}{149\,\Omega} = \boxed{0.080\ A}$

Note that $I = I_1 + I_2$

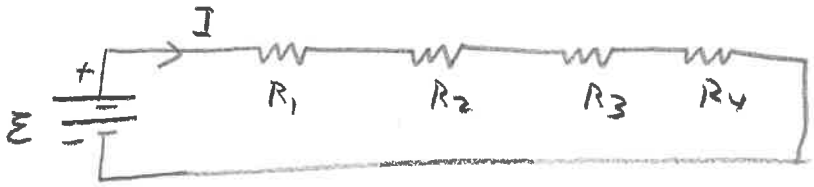
$I_1 = \frac{\varepsilon}{R_1} = \frac{12\ V}{250\,\Omega} = \boxed{0.048\ A}$

$I_2 = \frac{\varepsilon}{R_2} = \frac{12\ V}{370} = \boxed{0.032\ A}$

(55.) $R_1 = 12\,\Omega$, $R_2 = 15\,\Omega$, $R_3 = 20\,\Omega$, $R_4 = 35\,\Omega$

a) $R_{eq} = R_1 + R_2 + R_3 + R_4 = \boxed{82\,\Omega}$ (series combination)

b)



, $\varepsilon = 6.0$ volts

$I = \frac{\varepsilon}{R_{eq}} = \boxed{0.73\ amp}$

$\Delta V_1 = R_1 I = \boxed{0.88\ volt}$

$\Delta V_2 = R_2 I = \boxed{1.10\ volts}$

$\Delta V_3 = R_3 I = \boxed{1.46\ volts}$

$\Delta V_4 = R_4 I = \boxed{2.56\ volts}$

Note: $\Delta V_1 + \Delta V_2 + \Delta V_3 + \Delta V_4 = 6.0\ volt = \varepsilon$

(69.) Hair dryer: $P = 1750$ watts

$\Delta V = 120$ V

a) $I = \frac{P}{\Delta V} = \boxed{14.58 \text{ Amps}}$

b) Energy: $E = P \Delta t$, $\Delta t = 10 \text{ min} = 600 \text{ s}$

$= 1750 \text{ W} \times 600 \text{ s}$

$= \boxed{1.05 \times 10^6 \text{ J}} \times \left(\frac{1 \text{ kW.h}}{3.6 \times 10^6 \text{ Joules}} \right)$

$= \boxed{0.292 \text{ kWh}}$

c) cost per kWh $= c = \frac{15 \text{ ¢}}{\text{kW.hr}}$

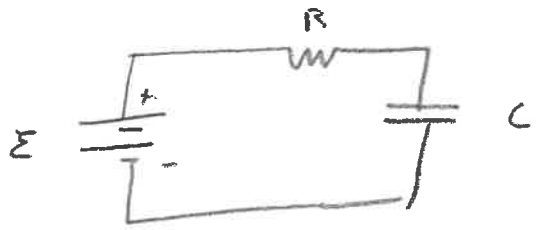
monthly cost:

$\text{monthly cost} = \frac{15 \text{ ¢}}{\text{kWh}} \times \left(\frac{0.292 \text{ kWh}}{\text{day}} \right) \times 30 \text{ day}$

$= 131 \text{ ¢}$

$= \boxed{\$1.31}$

(75.)



$\varepsilon = 6$ volts,

$C = 300 \text{ pF} = 300 \times 10^{-12} \text{ F}$

$R = 175 \text{ M}\Omega = 175 \times 10^6 \ \Omega$

$q(t) = q_{max}\left(1 - e^{-t/\tau}\right)$

where $q_{max} = C \cdot \varepsilon = \boxed{1.8 \text{ nC}}$, $\tau = RC = 0.0525 \text{ s} =$

a) $q(t = 1 \text{ ms}) = q_{max}\left(1 - e^{-1 \times 10^{-3} \text{s}/\tau}\right) = \boxed{0.034 \text{ nC}}$

b) $q(t = 10 \text{ ms}) = q_{max}\left(1 - e^{-10 \times 10^{-3} \text{s}/\tau}\right) = \boxed{0.312 \text{ nC}}$

c) $q(t = 100 \text{ ms}) = q_{max}\left(1 - e^{-100 \times 10^{-3} \text{s}/\tau}\right) = \boxed{1.53 \text{ nC}}$

(77.) $q(t) = q_{max}\left(1 - e^{-t/\tau}\right)$

$\tau = RC$, $q_{max} = \mathcal{E} C$

a) What is t/τ for $q(t) = \frac{1}{2} q_{max}$?

Ans: $\frac{1}{2} q_{max} = q_{max}\left(1 - e^{-t/\tau}\right)$

$\frac{1}{2} = 1 - e^{-t/\tau}$

$e^{-t/\tau} = 1 - \frac{1}{2} = \frac{1}{2}$

$\rightarrow \quad -t/\tau = \ln\left(\frac{1}{2}\right)$

$t/\tau = \ln(2) = \boxed{0.693}$

b) What is t/τ for $U(t) = \frac{1}{2} U_{max}$?

Recall: $U_{max} = \frac{1}{2} q_{max} \mathcal{E} = \frac{1}{2C} q_{max}^2$

$U(t) = \frac{1}{2C} q^2(t)$

Thus, $U(t) = \frac{1}{2} U_{max} \iff q^2(t) = \frac{1}{2} q_{max}^2$

$q(t) = \frac{1}{\sqrt{2}} q_{max}$

$\rightarrow \quad \frac{1}{\sqrt{2}} q_{max} = q_{max}\left(1 - e^{-t/\tau}\right)$

$\frac{1}{\sqrt{2}} = 1 - e^{-t/\tau}$

$e^{-t/\tau} = 1 - \frac{1}{\sqrt{2}}$

$\rightarrow \quad -t/\tau = \ln\left(1 - \frac{1}{\sqrt{2}}\right)$

$t/\tau = -\ln\left(1 - \frac{1}{\sqrt{2}}\right) = \boxed{1.23}$